LeafNet-BD: Preventing Mango Sapling Fraud Using Deep Learning and Leaf Image Classification

Abstract

Mango (Mangifera indica L.) is one of the most economically significant fruit crops in Bangladesh, with different varieties being grown to cater to different consumer preferences. Nevertheless, farmers and buyers are regularly left confused about authenticity while purchasing saplings as mislabeling or fraudulent sales of mango varieties are common. This paper introduces LeafNet-BD, which is a deep learning model designed specifically to detect different varieties of mangoes via leaf images. The proposed model has exploited MobileNetV2, which is an efficient CNN designed specifically to work in resource-constrained settings, such as mobile devices and field settings. The model was trained to classify the different mango leaf varieties with high accuracy using the MangoLeafVarietyBD dataset, with 2,744 raw images of eight different varieties. The model attained an accuracy around 90% and was further converted to work on web-based interaction via the Gradio tool. LeafNet-BD has proven to be an affordable deep learning method usable for authenticity verification with the potential to minimize fraud in the mango nursery industry in Bangladesh.

Index Terms

Mango variety identification; deep learning; MobileNetV2; agricultural authenticity; image classification

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